## 6.5 Positive Definite Matrices

This section concentrates on ***symmetric matrices that have positive eigenvalues.*** If symmetry makes a matrix important, this extra property (*all* $\lambda > 0$) makes it truly special. When we say special, we don't mean rare. Symmetric matrices with positive eigenvalues are at the center of all kinds of applications. They are called ***positive definite.***

The first problem is to recognize these matrices. You may say, just find the eigenvalues and test $\lambda > 0$. That is exactly what we want to avoid. Calculating eigenvalues is work. When the $\lambda$'s are needed, we can compute them. But if we just want to know that they are positive, there are faster ways. Here are two goals of this section:

- To find *quick tests* on a symmetric matrix that guarantee *positive eigenvalues.*
- To explain important applications of positive definiteness.

The $\lambda$'s are automatically real because the matrix is symmetric.

***Start with 2 by 2. When does*** $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ ***have*** $\lambda_1 > 0$ ***and*** $\lambda_2 > 0$?

***The eigenvalues of A are positive if and only if*** $a > 0$ ***and*** $ac - b^2 > 0$.

$A_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is *not* positive definite because $ac - b^2 = 1 - 4 < 0$

$A_2 = \begin{bmatrix} 1 & -2 \\ -2 & 6 \end{bmatrix}$ is positive definite because $a = 1$ and $ac - b^2 = 6 - 4 > 0$

$A_3 = \begin{bmatrix} -1 & 2 \\ 2 & -6 \end{bmatrix}$ is *not* positive definite (even with $\det A = +2$) because $a = -1$

Notice that we didn't compute the eigenvalues 3 and $-1$ of $A_1$. Positive trace $3 - 1 = 2$, negative determinant $(3)(-1) = -3$. And $A_3 = -A_2$ is *negative* definite. The positive eigenvalues for $A_2$, two negative eigenvalues for $A_3$.

*Proof that the 2 by 2 test is passed when* $\lambda_1 > 0$ *and* $\lambda_2 > 0$. Their product $\lambda_1\lambda_2$ is the determinant so $ac - b^2 > 0$. Their sum is the trace so $a + c > 0$. Then $a$ and $c$ are both positive (if one of them is not positive, $ac - b^2 > 0$ will fail). Problem 1 reverses the reasoning to show that the tests guarantee $\lambda_1 > 0$ and $\lambda_2 > 0$.

This test uses the 1 by 1 determinant $a$ and the 2 by 2 determinant $ac - b^2$. When $A$ is 3 by 3, $\det A > 0$ is the third part of the test. The next test requires *positive pivots.*

***The eigenvalues of*** $A = A^{\mathrm{T}}$ ***are positive if and only if the pivots are positive:***

$$a > 0 \quad \text{and} \quad \frac{ac - b^2}{a} > 0.$$

$a > 0$ is required in both tests. So $ac > b^2$ is also required, for the determinant test and now the pivot. The point is to recognize that ratio as the *second pivot* of $A$:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \begin{matrix} \text{The first pivot is } a \\ \longrightarrow \\ \text{The multiplier is } b/a \end{matrix} \quad \begin{bmatrix} a & b \\ 0 & c - \frac{b}{a}b \end{bmatrix} \quad \textit{The second pivot is } c - \frac{b^2}{a} = \frac{ac - b^2}{a}$$

This connects two big parts of linear algebra. ***Positive eigenvalues mean positive pivots and vice versa.*** We gave a proof for symmetric matrices of any size in the last section. The pivots give a quick test for $\lambda > 0$, and they are a lot faster to compute than the eigenvalues. It is very satisfying to see pivots and determinants and eigenvalues come together in this course.

$$A_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \qquad A_2 = \begin{bmatrix} 1 & -2 \\ -2 & 6 \end{bmatrix} \qquad A_3 = \begin{bmatrix} -1 & 2 \\ 2 & -6 \end{bmatrix}$$

pivots 1 and $-3$ **(indefinite)** | pivots 1 and 2 **(positive definite)** | pivots $-1$ and $-2$ **(negative definite)**

Here is a different way to look at symmetric matrices with positive eigenvalues.

## Energy-based Definition

From $Ax = \lambda x$, multiply by $x^T$ to get $x^TAx = \lambda x^Tx$. The right side is a positive $\lambda$ times a positive number $x^Tx = \|x\|^2$. So $x^TAx$ is positive for any eigenvector.

The new idea is that ***$x^TAx$ is positive for all nonzero vectors $x$***, not just the eigenvectors. In many applications this number $x^TAx$ (or $\frac{1}{2}x^TAx$) is the **energy** in the system. The requirement of positive energy gives *another definition* of a positive definite matrix. I think this energy-based definition is the fundamental one.

Eigenvalues and pivots are two equivalent ways to test the new requirement $x^TAx > 0$.

**Definition** ***$A$ is positive definite if $x^TAx > 0$ for every nonzero vector $x$:***

$$x^TAx = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = ax^2 + 2bxy + cy^2 > 0. \tag{1}$$

The four entries $a, b, b, c$ give the four parts of $x^TAx$. From $a$ and $c$ come the pure squares $ax^2$ and $cy^2$. From $b$ and $b$ off the diagonal come the cross terms $bxy$ and $byx$ (the same). Adding those four parts gives $x^TAx$. This energy-based definition leads to a basic fact:

*If $A$ and $B$ are symmetric positive definite, so is $A + B$.*

**Reason:** $x^T(A + B)x$ is simply $x^TAx + x^TBx$. Those two terms are positive (for $x \neq 0$) so $A + B$ is also positive definite. The pivots and eigenvalues are not easy to follow when matrices are added, but the energies just add.

$x^T Ax$ also connects with our final way to recognize a positive definite matrix. Start with any matrix $R$, possibly rectangular. We know that $A = R^T R$ is square and symmetric. More than that, $A$ will be positive definite when $R$ has independent columns:

*If the columns of $R$ are independent, then $A = R^T R$ is positive definite.*

Again eigenvalues and pivots are not easy. But the number $x^T Ax$ is the same as $x^T R^T Rx$. That is exactly $(Rx)^T(Rx)$—another important proof by parenthesis! That vector $Rx$ is not zero when $x \neq 0$ (this is the meaning of independent columns). Then $x^T Ax$ is the positive number $\|Rx\|^2$ and the matrix $A$ is positive definite.

Let me collect this theory together, into five equivalent statements of positive definiteness. You will see how that key idea connects the whole subject of linear algebra: pivots, determinants, eigenvalues, and least squares (from $R^T R$). Then come the applications.

***When a symmetric matrix has one of these five properties, it has them all:***

1. All ***n pivots*** are positive.
2. All ***n upper left determinants*** are positive.
3. All ***n eigenvalues*** are positive.
4. $x^T Ax$ is positive except at $x = 0$. This is the ***energy-based*** definition.
5. $A$ equals $R^T R$ for a matrix $R$ with ***independent columns.***

The "upper left determinants" are 1 by 1, 2 by 2, . . ., $n$ by $n$. The last one is the determinant of the complete matrix $A$. This remarkable theorem ties together the whole linear algebra course—at least for symmetric matrices. We believe that two examples are more helpful than a detailed proof (we nearly have a proof already).

**Example 1** Test these matrices $A$ and $B$ for positive definiteness:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 & b \\ -1 & 2 & -1 \\ b & -1 & 2 \end{bmatrix}.$$

**Solution** The pivots of $A$ are 2 and $\frac{3}{2}$ and $\frac{4}{3}$, all positive. Its upper left determinants are 2 and 3 and 4, all positive. The eigenvalues of $A$ are $2 - \sqrt{2}$ and 2 and $2 + \sqrt{2}$, all positive. That completes tests **1**, **2**, and **3**.

We can write $x^T Ax$ as a sum of three squares. The pivots 2, $\frac{3}{2}$, $\frac{4}{3}$ appear outside the squares. The multipliers $-\frac{1}{2}$ and $-\frac{2}{3}$ from elimination are inside the squares:

$$x^T Ax = 2(x_1^2 - x_1x_2 + x_2^2 - x_2x_3 + x_3^2) \qquad \textbf{Rewrite with squares}$$

$$= 2(x_1 - \tfrac{1}{2}x_2)^2 + \tfrac{3}{2}(x_2 - \tfrac{2}{3}x_3)^2 + \tfrac{4}{3}(x_3)^2. \qquad \textbf{This sum is positive.}$$

I have two candidates to suggest for $R$. Either one will show that $A = R^{\mathrm{T}}R$ is positive definite. $R$ can be a rectangular first difference matrix, 4 by 3, to produce those second differences $-1, 2, -1$ in $A$:

$$A = R^{\mathrm{T}}R \quad \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

The three columns of this $R$ are independent. $A$ is positive definite.

Another $R$ comes from $A = LDL^{\mathrm{T}}$ (the symmetric version of $A = LU$). Elimination gives the pivots $2, \frac{3}{2}, \frac{4}{3}$ in $D$ and the multipliers $-\frac{1}{2}, 0, -\frac{2}{3}$ in $L$. **Just put $\sqrt{D}$ with $L$.**

$$LDL^{\mathrm{T}} = \begin{bmatrix} 1 & & \\ -\frac{1}{2} & 1 & \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & \frac{3}{2} & \\ & & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & \\ & 1 & -\frac{2}{3} \\ & & 1 \end{bmatrix} = (L\sqrt{D})(L\sqrt{D})^{\mathrm{T}} = R^{\mathrm{T}}R. \quad (2)$$

***R is the Cholesky factor***

This choice of $R$ has square roots (not so beautiful). But it is the only $R$ that is 3 by 3 and upper triangular. It is the "Cholesky factor" of $A$ and it is computed by MATLAB's command $R = \mathrm{chol}(A)$. In applications, the rectangular $R$ is how we build $A$ and this Cholesky $R$ is how we break it apart.

Eigenvalues give the symmetric choice $R = Q\sqrt{\Lambda}Q^{\mathrm{T}}$. This is also successful with $R^{\mathrm{T}}R = Q\Lambda Q^{\mathrm{T}} = A$. All these tests show that the $-1, 2, -1$ matrix $A$ is positive definite.

Now turn to $B$, where the $(1,3)$ and $(3,1)$ entries move away from 0 to $b$. This $b$ must not be too large! *The determinant test is easiest.* The 1 by 1 determinant is 2, the 2 by 2 determinant is still 3. The 3 by 3 determinant involves $b$:

$$\det B = 4 + 2b - 2b^2 = (1 + b)(4 - 2b) \quad \text{must be positive.}$$

At $b = -1$ and $b = 2$ we get $\det B = 0$. *Between $b = -1$ and $b = 2$ the matrix is positive definite.* The corner entry $b = 0$ in the first matrix $A$ was safely between.

## Positive Semidefinite Matrices

Often we are at the edge of positive definiteness. The determinant is zero. The smallest eigenvalue is zero. The energy in its eigenvector is $x^{\mathrm{T}}Ax = x^{\mathrm{T}}0x = 0$. These matrices on the edge are called *positive semidefinite*. Here are two examples (not invertible):

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \text{ are positive semidefinite.}$$

$A$ has eigenvalues 5 and 0. Its upper left determinants are 1 and 0. Its rank is only 1. This matrix $A$ factors into $R^{\mathrm{T}}R$ with **dependent columns** in $R$:

**Dependent columns**
**Positive semidefinite**

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = R^{\mathrm{T}}R.$$

If 4 is increased by any small number, the matrix will become positive definite.

The cyclic $B$ also has zero determinant (computed above when $b = -1$). It is singular. The eigenvector $\boldsymbol{x} = (1, 1, 1)$ has $B\boldsymbol{x} = \mathbf{0}$ and $\boldsymbol{x}^{\mathrm{T}}B\boldsymbol{x} = 0$. Vectors $\boldsymbol{x}$ in all other directions do give positive energy. This $B$ can be written as $R^{\mathrm{T}}R$ in many ways, but $R$ will always have *dependent* columns, with $(1, 1, 1)$ in its nullspace:

**Second differences $A$ from first differences $R^{\mathrm{T}}R$ Cyclic $A$ from cyclic $R$**

$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

Positive semidefinite matrices have all $\lambda \geq 0$ and all $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} \geq 0$. Those weak inequalities ($\geq$ instead of $>$) include positive definite matrices and the singular matrices at the edge.

## First Application: The Ellipse $ax^2 + 2bxy + cy^2 = 1$

Think of a tilted ellipse $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = 1$. Its center is $(0, 0)$, as in Figure 6.7a. Turn it to line up with the coordinate axes ($X$ and $Y$ axes). That is Figure 6.7b. These two pictures show the geometry behind the factorization $A = Q\Lambda Q^{-1} = Q\Lambda Q^{\mathrm{T}}$:

1. The tilted ellipse is associated with $A$. Its equation is $\boldsymbol{x}^{\mathrm{T}}A\boldsymbol{x} = 1$.
2. The lined-up ellipse is associated with $\Lambda$. Its equation is $\boldsymbol{X}^{\mathrm{T}}\Lambda\boldsymbol{X} = 1$.
3. The rotation matrix that lines up the ellipse is the eigenvector matrix $Q$.

**Example 2** Find the axes of this tilted ellipse $5x^2 + 8xy + 5y^2 = 1$.

**Solution** Start with the positive definite matrix that matches this equation:

The equation is $\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 1.$ The matrix is $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$

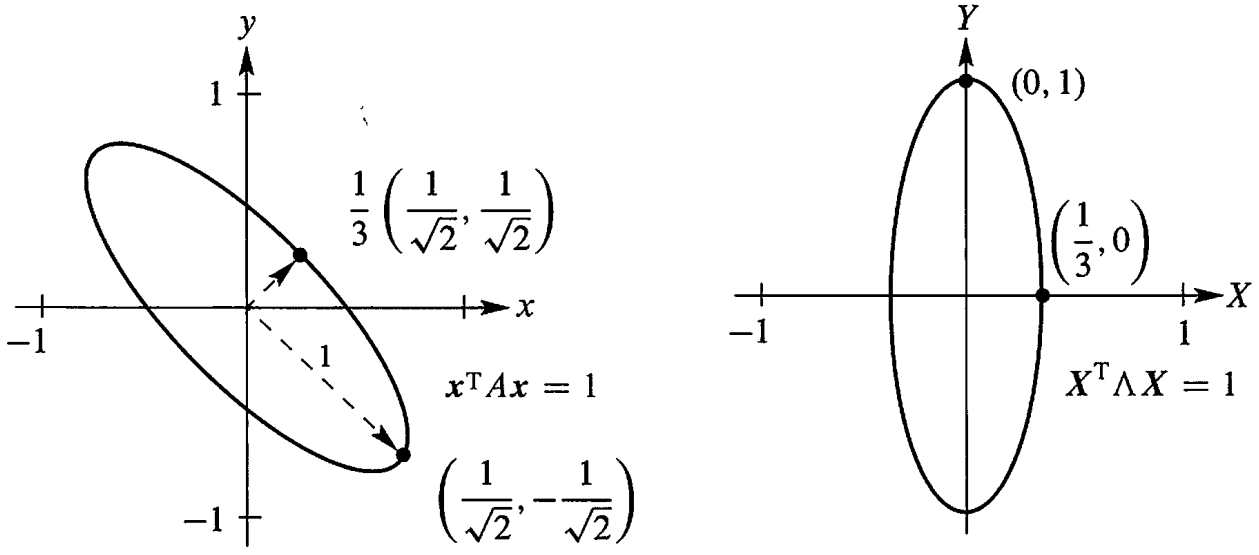



Figure 6.7: The tilted ellipse $5x^2 + 8xy + 5y^2 = 1$. Lined up it is $9X^2 + Y^2 = 1$.

The eigenvectors are $\begin{bmatrix}1\\1\end{bmatrix}$ and $\begin{bmatrix}-1\\1\end{bmatrix}$. Divide by $\sqrt{2}$ for unit vectors. Then $A = Q\Lambda Q^{\mathrm{T}}$:

**Eigenvectors in $Q$**
**Eigenvalues 9 and 1**

$$\begin{bmatrix}5 & 4\\4 & 5\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}9 & 0\\0 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}.$$

Now multiply by $\begin{bmatrix}x & y\end{bmatrix}$ on the left and $\begin{bmatrix}x\\y\end{bmatrix}$ on the right to get back to $x^{\mathrm{T}}Ax$:

$$x^{\mathrm{T}}Ax = \textbf{sum of squares} \quad 5x^2 + 8xy + 5y^2 = 9\left(\frac{x+y}{\sqrt{2}}\right)^2 + 1\left(\frac{x-y}{\sqrt{2}}\right)^2. \tag{3}$$

The coefficients are not the pivots 5 and 9/5 from $D$, they are the eigenvalues 9 and 1 from $\Lambda$. Inside *these* squares are the eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$.

***The axes of the tilted ellipse point along the eigenvectors.*** This explains why $A = Q\Lambda Q^{\mathrm{T}}$ is called the "principal axis theorem"—it displays the axes. Not only the axis directions (from the eigenvectors) but also the axis lengths (from the eigenvalues). To see it all, use capital letters for the new coordinates that line up the ellipse:

$$\textbf{Lined up} \qquad \frac{x+y}{\sqrt{2}} = X \quad \text{and} \quad \frac{x-y}{\sqrt{2}} = Y \quad \text{and} \quad 9X^2 + Y^2 = 1.$$

The largest value of $X^2$ is $1/9$. The endpoint of the shorter axis has $X = 1/3$ and $Y = 0$. Notice: The *bigger* eigenvalue $\lambda_1$ gives the *shorter* axis, of half-length $1/\sqrt{\lambda_1} = 1/3$. The smaller eigenvalue $\lambda_2 = 1$ gives the greater length $1/\sqrt{\lambda_2} = 1$.

In the $xy$ system, the axes are along the eigenvectors of $A$. In the $XY$ system, the axes are along the eigenvectors of $\Lambda$—the coordinate axes. All comes from $A = Q\Lambda Q^{\mathrm{T}}$.

Suppose $A = Q\Lambda Q^{\mathrm{T}}$ is positive definite, so $\lambda_i > 0$. The graph of $x^{\mathrm{T}}Ax = 1$ is an ellipse:

$$\begin{bmatrix}x & y\end{bmatrix} Q\Lambda Q^{\mathrm{T}}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}X & Y\end{bmatrix}\Lambda\begin{bmatrix}X\\Y\end{bmatrix} = \lambda_1 X^2 + \lambda_2 Y^2 = 1.$$

The axes point along eigenvectors. The half-lengths are $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$.

$A = I$ gives the circle $x^2 + y^2 = 1$. If one eigenvalue is negative (exchange 4's and 5's in $A$), we don't have an ellipse. The sum of squares becomes a *difference of squares*: $9X^2 - Y^2 = 1$. This indefinite matrix gives a *hyperbola*. For a negative definite matrix like $A = -I$, with both $\lambda$'s negative, the graph of $-x^2 - y^2 = 1$ has no points at all.

## ■ REVIEW OF THE KEY IDEAS ■

1. Positive definite matrices have positive eigenvalues and positive pivots.
2. A quick test is given by the upper left determinants: $a > 0$ and $ac - b^2 > 0$.

3. The graph of $x^T Ax$ is then a "bowl" going up from $x = 0$:

$$x^T Ax = ax^2 + 2bxy + cy^2 \text{ is positive except at } (x, y) = (0, 0).$$

4. $A = R^T R$ is automatically positive definite if $R$ has independent columns.

5. The ellipse $x^T Ax = 1$ has its axes along the eigenvectors of $A$. Lengths $1/\sqrt{\lambda}$.

## ■ WORKED EXAMPLES ■

**6.5 A** The great factorizations of a symmetric matrix are $A = LDL^T$ from pivots and multipliers, and $A = Q\Lambda Q^T$ from eigenvalues and eigenvectors. Show that $x^T Ax > 0$ for all nonzero $x$ exactly when the pivots and eigenvalues are positive. Try these $n$ by $n$ tests on pascal(6) and ones(6) and hilb(6) and other matrices in MATLAB's gallery.

**Solution** To prove $x^T Ax > 0$, put parentheses into $x^T LDL^T x$ and $x^T Q\Lambda Q^T x$:

$$x^T Ax = (L^T x)^T D(L^T x) \qquad \text{and} \qquad x^T Ax = (Q^T x)^T \Lambda (Q^T x).$$

If $x$ is nonzero, then $y = L^T x$ and $z = Q^T x$ are nonzero (those matrices are invertible). So $x^T Ax = y^T Dy = z^T \Lambda z$ becomes a sum of squares and $A$ is shown as positive definite:

| | | | | | | |
|---|---|---|---|---|---|---|
| **Pivots** | $x^T Ax$ | $=$ | $y^T Dy$ | $=$ | $d_1 y_1^2 + \cdots + d_n y_n^2$ | $> 0$ |
| **Eigenvalues** | $x^T Ax$ | $=$ | $z^T \Lambda z$ | $=$ | $\lambda_1 z_1^2 + \cdots + \lambda_n z_n^2$ | $> 0$ |

MATLAB has a gallery of unusual matrices (type help gallery) and here are four:

**pascal(6)** is positive definite because all its pivots are 1 (Worked Example **2.6 A**).

**ones(6)** is positive *semidefinite* because its eigenvalues are $0, 0, 0, 0, 0, 6$.

**H=hilb(6)** is positive definite even though eig(H) shows two eigenvalues very near zero.

**Hilbert matrix** $x^T Hx = \int_0^1 (x_1 + x_2 s + \cdots + x_6 s^5)^2 \, ds > 0$, $H_{ij} = 1/(i + j + 1)$.

**rand(6)+rand(6)'** can be positive definite or not. Experiments gave only 2 in 20000.

$n = 20000$; $p = 0$; for $k = 1{:}n$, $A =$ rand(6); $p = p +$ all(eig($A + A'$) $> 0$); end, $p / n$

**6.5 B When is the symmetric block matrix** $M = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$ **positive definite?**

**Solution** Multiply the first row of $M$ by $B^T A^{-1}$ and subtract from the second row, to get a block of zeros. The *Schur complement* $S = C - B^T A^{-1} B$ appears in the corner:

$$\begin{bmatrix} I & 0 \\ -B^T A^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ B^T & C \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & C - B^T A^{-1} B \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \tag{4}$$

***Those two blocks A and S must be positive definite.*** Their pivots are the pivots of $M$.

**6.5 C** *Second application: Test for a minimum.* Does $F(x, y)$ have a minimum if $\partial F/\partial x = 0$ and $\partial F/\partial y = 0$ at the point $(x, y) = (0, 0)$?

**Solution** For $f(x)$, the test for a minimum comes from calculus: $df/dx = 0$ and $d^2f/dx^2 > 0$. Moving to two variables $x$ and $y$ produces a symmetric matrix $H$. It contains the four second derivatives of $F(x, y)$. Positive $f''$ changes to positive definite $H$:

**Second derivative matrix**
$$H = \begin{bmatrix} \partial^2 F/\partial x^2 & \partial^2 F/\partial x \partial y \\ \partial^2 F/\partial y \partial x & \partial^2 F/\partial y^2 \end{bmatrix}$$

***$F(x, y)$ has a minimum if $H$ is positive definite.*** Reason: $H$ reveals the important terms $ax^2 + 2bxy + cy^2$ near $(x, y) = (0, 0)$. The second derivatives of $F$ are $2a, 2b, 2b, 2c$!

**6.5 D** Find the eigenvalues of the $-1, 2, -1$ tridiagonal $n$ by $n$ matrix $K$ (my favorite).

**Solution** The best way is to guess $\lambda$ and $\boldsymbol{x}$. Then check $K\boldsymbol{x} = \lambda \boldsymbol{x}$. Guessing could not work for most matrices, but special cases are a big part of mathematics (pure and applied).

The key is hidden in a differential equation. The second difference matrix $K$ is like a *second derivative*, and those eigenvalues are much easier to see:

**Eigenvalues** $\lambda_1, \lambda_2, \ldots$
**Eigenfunctions** $y_1, y_2, \ldots$
$$\frac{d^2y}{dx^2} = \lambda y(x) \quad \textbf{with} \quad \begin{matrix} y(0) = 0 \\ y(1) = 0 \end{matrix} \tag{5}$$

Try $y = \sin cx$. Its second derivative is $y'' = -c^2 \sin cx$. So the eigenvalue will be $\lambda = -c^2$, provided $y(x)$ satisfies the end point conditions $y(0) = 0 = y(1)$.

Certainly $\sin 0 = 0$ (this is where cosines are eliminated by $\cos 0 = 1$). At $x = 1$, we need $y(1) = \sin c = 0$. The number $c$ must be $k\pi$, a multiple of $\pi$, and $\lambda$ is $-c^2$:

**Eigenvalues** $\lambda = -k^2\pi^2$
**Eigenfunctions** $y = \sin k\pi x$
$$\frac{d^2}{dx^2} \sin k\pi x = -k^2\pi^2 \sin k\pi x. \tag{6}$$

Now we go back to the matrix $K$ and guess its eigenvectors. They come from $\sin k\pi x$ at $n$ points $x = h, 2h, \ldots, nh$, equally spaced between 0 and 1. The spacing $\Delta x$ is $h = 1/(n + 1)$, so the $(n + 1)$st point comes out at $(n + 1)h = 1$. Multiply that sine vector $\boldsymbol{s}$ by $K$:

**Eigenvector of $K$ = sine vector $\boldsymbol{s}$**
$$\begin{aligned} K\boldsymbol{s} &= \lambda \boldsymbol{s} = (2 - 2\cos k\pi h)\, \boldsymbol{s} \\ \boldsymbol{s} &= (\sin k\pi h, \ldots, \sin nk\pi h). \end{aligned} \tag{7}$$

I will leave that multiplication $K\boldsymbol{s} = \lambda \boldsymbol{s}$ as a challenge problem. Notice what is important:

1. All eigenvalues $2 - 2\cos k\pi h$ are positive and $K$ is positive definite.

2. The **sine matrix** $S$ has orthogonal columns = eigenvectors $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_n$ of $K$.

**Discrete Sine Transform**
**The $j,k$ entry is** $\sin jk\pi h$

$$S = \begin{bmatrix} \sin \pi h & & \sin k\pi h & \\ \vdots & \cdots & \vdots & \cdots \\ \sin n\pi h & & \sin nk\pi h & \end{bmatrix}$$

Those eigenvectors are orthogonal just like the eigenfunctions: $\int_0^1 \sin j\pi x \sin k\pi x \, dx = 0$.

# Problem Set 6.5

**Problems 1–13 are about tests for positive definiteness.**

**1** Suppose the 2 by 2 tests $a > 0$ and $ac - b^2 > 0$ are passed. Then $c > b^2/a$ is also positive.

(i) $\lambda_1$ and $\lambda_2$ have the *same sign* because their product $\lambda_1\lambda_2$ equals _____.

(i) That sign is positive because $\lambda_1 + \lambda_2$ equals _____.

*Conclusion*: The tests $a > 0, ac - b^2 > 0$ guarantee positive eigenvalues $\lambda_1, \lambda_2$.

**2** Which of $A_1, A_2, A_3, A_4$ has two positive eigenvalues? Use the test, don't compute the $\lambda$'s. Find an $x$ so that $x^{\mathrm{T}}A_1x < 0$, so $A_1$ fails the test.

$$A_1 = \begin{bmatrix} 5 & 6 \\ 6 & 7 \end{bmatrix} \quad A_2 = \begin{bmatrix} -1 & -2 \\ -2 & -5 \end{bmatrix} \quad A_3 = \begin{bmatrix} 1 & 10 \\ 10 & 100 \end{bmatrix} \quad A_4 = \begin{bmatrix} 1 & 10 \\ 10 & 101 \end{bmatrix}.$$

**3** For which numbers $b$ and $c$ are these matrices positive definite?

$$A = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} \qquad A = \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \qquad A = \begin{bmatrix} c & b \\ b & c \end{bmatrix}.$$

With the pivots in $D$ and multiplier in $L$, factor each $A$ into $LDL^{\mathrm{T}}$.

**4** What is the quadratic $f = ax^2 + 2bxy + cy^2$ for each of these matrices? Complete the square to write $f$ as a sum of one or two squares $d_1(\ \ )^2 + d_2(\ \ )^2$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 9 \end{bmatrix} \qquad \text{and} \qquad A = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}.$$

**5** Write $f(x, y) = x^2 + 4xy + 3y^2$ as a *difference* of squares and find a point $(x, y)$ where $f$ is negative. The minimum is not at $(0, 0)$ even though $f$ has positive coefficients.

**6** The function $f(x, y) = 2xy$ certainly has a saddle point and not a minimum at $(0, 0)$. What symmetric matrix $A$ produces this $f$? What are its eigenvalues?

**7** Test to see if $R^TR$ is positive definite in each case:

$$R = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

**8** The function $f(x, y) = 3(x + 2y)^2 + 4y^2$ is positive except at $(0, 0)$. What is the matrix in $f = [x \quad y]A[x \quad y]^T$? Check that the pivots of $A$ are 3 and 4.

**9** Find the 3 by 3 matrix $A$ and its pivots, rank, eigenvalues, and determinant:

$$[x_1 \quad x_2 \quad x_3] \begin{bmatrix} \quad & \quad & \quad \\ & A & \\ \quad & \quad & \quad \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4(x_1 - x_2 + 2x_3)^2.$$

**10** Which 3 by 3 symmetric matrices $A$ and $B$ produce these quadratics?

$x^TAx = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3)$. Why is $A$ positive definite?

$x^TBx = 2(x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_1x_3 - x_2x_3)$. Why is $B$ semidefinite?

**11** Compute the three upper left determinants of $A$ to establish positive definiteness. Verify that their ratios give the second and third pivots.

**Pivots = ratios of determinants** $\qquad A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix}.$

**12** For what numbers $c$ and $d$ are $A$ and $B$ positive definite? Test the 3 determinants:

$$A = \begin{bmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & d & 4 \\ 3 & 4 & 5 \end{bmatrix}.$$

**13** Find a matrix with $a > 0$ and $c > 0$ and $a + c > 2b$ that has a negative eigenvalue.

**Problems 14–20 are about applications of the tests.**

**14** *If $A$ is positive definite then $A^{-1}$ is positive definite.* Best proof: The eigenvalues of $A^{-1}$ are positive because _____. *Second proof* (only for 2 by 2):

The entries of $A^{-1} = \dfrac{1}{ac - b^2}\begin{bmatrix} c & -b \\ -b & a \end{bmatrix}$ pass the determinant tests _____.

**15** *If $A$ and $B$ are positive definite, their sum $A + B$ is positive definite.* Pivots and eigenvalues are not convenient for $A + B$. Better to prove $x^T(A + B)x > 0$. Or if $A = R^TR$ and $B = S^TS$, show that $A + B = [R \; S]^T \begin{bmatrix} R \\ S \end{bmatrix}$ with independent columns.

**16** A positive definite matrix cannot have a zero (or even worse, a negative number) on its diagonal. Show that this matrix fails to have $x^{\mathrm{T}}Ax > 0$:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 4 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \text{ is not positive when } (x_1, x_2, x_3) = (\ \ ,\ \ ,\ \ ).$$

**17** A diagonal entry $a_{jj}$ of a symmetric matrix cannot be smaller than all the $\lambda$'s. If it were, then $A - a_{jj}I$ would have _____ eigenvalues and would be positive definite. But $A - a_{jj}I$ has a _____ on the main diagonal.

**18** If $Ax = \lambda x$ then $x^{\mathrm{T}}Ax =$ _____. If $x^{\mathrm{T}}Ax > 0$, prove that $\lambda > 0$.

**19** Reverse Problem 18 to show that *if all* $\lambda > 0$ *then* $x^{\mathrm{T}}Ax > 0$. We must do this for *every* nonzero $x$, not just the eigenvectors. So write $x$ as a combination of the eigenvectors and *explain why all "cross terms" are* $x_i^{\mathrm{T}}x_j = 0$. Then $x^{\mathrm{T}}Ax$ is

$$(c_1x_1 + \cdots + c_nx_n)^{\mathrm{T}}(c_1\lambda_1x_1 + \cdots + c_n\lambda_nx_n) = c_1^2\lambda_1x_1^{\mathrm{T}}x_1 + \cdots + c_n^2\lambda_nx_n^{\mathrm{T}}x_n > 0.$$

**20** Give a quick reason why each of these statements is true:

(a) Every positive definite matrix is invertible.

(b) The only positive definite projection matrix is $P = I$.

(c) A diagonal matrix with positive diagonal entries is positive definite.

(d) A symmetric matrix with a positive determinant might not be positive definite!

**Problems 21–24 use the eigenvalues; Problems 25–27 are based on pivots.**

**21** For which $s$ and $t$ do $A$ and $B$ have all $\lambda > 0$ (therefore positive definite)?

$$A = \begin{bmatrix} s & -4 & -4 \\ -4 & s & -4 \\ -4 & -4 & s \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} t & 3 & 0 \\ 3 & t & 4 \\ 0 & 4 & t \end{bmatrix}.$$

**22** From $A = Q\Lambda Q^{\mathrm{T}}$ compute the positive definite symmetric square root $Q\Lambda^{1/2}Q^{\mathrm{T}}$ of each matrix. Check that this square root gives $R^2 = A$:

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 10 & 6 \\ 6 & 10 \end{bmatrix}.$$

**23** You may have seen the equation for an ellipse as $x^2/a^2 + y^2/b^2 = 1$. What are $a$ and $b$ when the equation is written $\lambda_1x^2 + \lambda_2y^2 = 1$? The ellipse $9x^2 + 4y^2 = 1$ has axes with half-lengths $a =$ _____ and $b =$ _____.

**24** Draw the tilted ellipse $x^2 + xy + y^2 = 1$ and find the half-lengths of its axes from the eigenvalues of the corresponding matrix $A$.

**25** With positive pivots in $D$, the factorization $A = LDL^T$ becomes $L\sqrt{D}\sqrt{D}L^T$. (Square roots of the pivots give $D = \sqrt{D}\sqrt{D}$.) Then $C = \sqrt{D}L^T$ yields the ***Cholesky factorization*** $A = C^TC$ which is "symmetrized $LU$":

$$\text{From } C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \text{ find } A. \qquad \text{From } A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \text{ find } C = \textbf{chol}(A).$$

**26** In the Cholesky factorization $A = C^TC$, with $C^T = L\sqrt{D}$, the square roots of the pivots are on the diagonal of $C$. Find $C$ (upper triangular) for

$$A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 7 \end{bmatrix}.$$

**27** The symmetric factorization $A = LDL^T$ means that $x^TAx = x^TLDL^Tx$:

$$\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix}\begin{bmatrix} a & 0 \\ 0 & (ac-b^2)/a \end{bmatrix}\begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}.$$

The left side is $ax^2 + 2bxy + cy^2$. The right side is $a\left(x + \frac{b}{a}y\right)^2 + ____y^2$. The second pivot completes the square! Test with $a = 2, b = 4, c = 10$.

**28** Without multiplying $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, find

(a) the determinant of $A$ (b) the eigenvalues of $A$
(c) the eigenvectors of $A$ (d) a reason why $A$ is symmetric positive definite.

**29** For $F_1(x,y) = \frac{1}{4}x^4 + x^2y + y^2$ and $F_2(x,y) = x^3 + xy - x$ find the second derivative matrices $H_1$ and $H_2$:

$$\textbf{Test for minimum: } H = \begin{bmatrix} \partial^2F/\partial x^2 & \partial^2F/\partial x\partial y \\ \partial^2F/\partial y\partial x & \partial^2F/\partial y^2 \end{bmatrix} \text{ is positive definite}$$

$H_1$ is positive definite so $F_1$ is concave up (= convex). Find the minimum point of $F_1$ and the saddle point of $F_2$ (look only where first derivatives are zero).

**30** The graph of $z = x^2 + y^2$ is a bowl opening upward. *The graph of $z = x^2 - y^2$ is a saddle.* The graph of $z = -x^2 - y^2$ is a bowl opening downward. What is a test on $a, b, c$ for $z = ax^2 + 2bxy + cy^2$ to have a saddle point at $(0,0)$?

**31** Which values of $c$ give a bowl and which $c$ give a saddle point for the graph of $z = 4x^2 + 12xy + cy^2$? Describe this graph at the borderline value of $c$.

## Challenge Problems

**32** A *group* of nonsingular matrices includes $AB$ and $A^{-1}$ if it includes $A$ and $B$. "Products and inverses stay in the group." Which of these are groups (as in 2.7.37)?

Invent a "subgroup" of two of these groups (not $I$ by itself $=$ the smallest group).

(a) Positive definite symmetric matrices $A$.

(b) Orthogonal matrices $Q$.

(c) All exponentials $e^{tA}$ of a fixed matrix $A$.

(d) Matrices $P$ with positive eigenvalues.

(e) Matrices $D$ with determinant 1.

**33** When $A$ and $B$ are symmetric positive definite, $AB$ might not even be symmetric. But its eigenvalues are still positive. Start from $ABx = \lambda x$ and take dot products with $Bx$. Then prove $\lambda > 0$.

**34** Write down the 5 by 5 sine matrix $S$ from Worked Example **6.5 D**, containing the eigenvectors of $K$ when $n = 5$ and $h = 1/6$. Multiply $K$ times $S$ to see the five positive eigenvalues.

Their sum should equal the trace 10. Their product should be $\det K = 6$.

**35** Suppose $C$ is positive definite (so $y^{\mathrm{T}}Cy > 0$ whenever $y \neq \mathbf{0}$) and $A$ has independent columns (so $Ax \neq \mathbf{0}$ whenever $x \neq \mathbf{0}$). Apply the energy test to $x^{\mathrm{T}}A^{\mathrm{T}}CAx$ to show that $A^{\mathrm{T}}CA$ is positive definite: *the crucial matrix in engineering.*